

ANNEXE M – DÉPLOIEMENT DOCKER ET RENDER

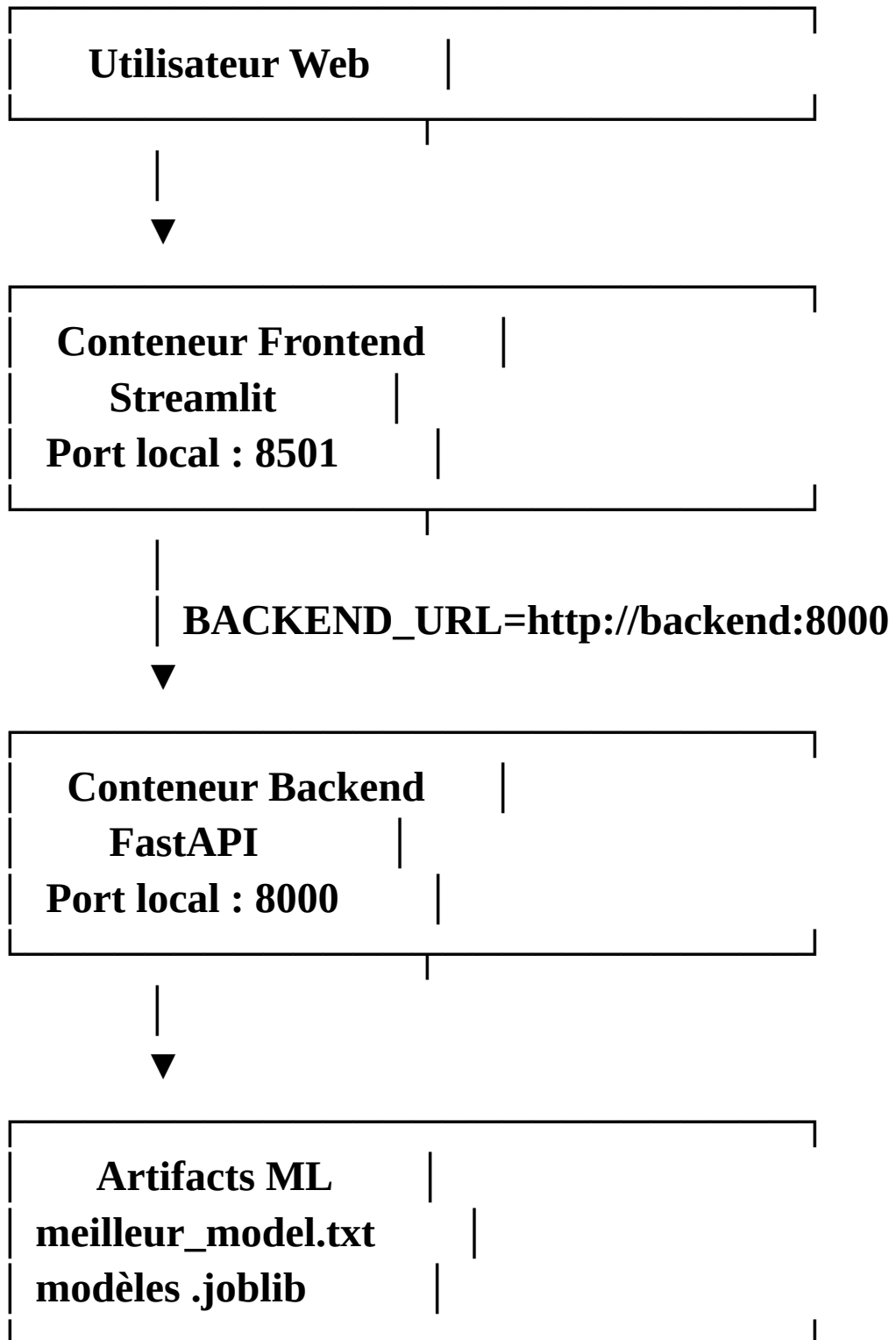
Conteneurisation

L'application repose sur :

- un conteneur Backend FastAPI ;
- un conteneur Frontend Streamlit.

Figure 5.1

— train_model.py	# Script principal : entraînement, évaluation, etc ...
— comparatif_model.py	# Graphique comparatif des performances des modèles
— Docker-compose.yml	# Orchestration backend + frontend
— requirements.txt	# Dépendances globales
— backend/	# API FastAPI pour la prédiction
— backend.py	# Endpoints /predict et /health
— Dockerfile	# Image Docker backend
— requirements.txt	# Dépendances backend
— frontend/	# Interface utilisateur Streamlit
— frontend.py	# UI : upload, prédiction, visualisation
— Dockerfile	# Image Docker frontend
— requirements.txt	# Dépendances frontend
— src/	# Code métier (ML + preprocessing)
— config.py	# Constantes globales (ARTIFACTS, TARGET, FEATURES...)
— data_loader.py	# Chargement des données depuis Supabase
— pipelines.py	# Pipelines ML (scaling, encodage...)
— train.py	# Fonctions d'entraînement
— transformation.py	# Feature engineering / preprocessing
— save.py	# Sauvegarde des modèles et métriques
— artifacts/	# Artefacts générés automatiquement
— *_metrics.json	# Métriques des modèles
— *_confusion_matrix.png	# Matrices de confusion
— *_learning_curve.png	# Courbes d'apprentissage
— *.joblib	# Modèles ML sérialisés
— meilleur_model.txt	# Nom du meilleur modèle sélectionné
— README.md	# Documentation du projet



Déploiement

Les services sont hébergés sur Render.

Figure 5.2

dashboard.render.com/web/srv-d8cru6k1jcs73a4h6vg/deployments/dep-d8s2cm0tmc739a93j0

My project / Production / Alz_indu_backend

Q Sea

WEB SERVICE

Alz_indu_backend

Docker Free Upgrade your instance →

Service ID: srv-d8cru6k1jcs73a4h6vg

zcesarini / Alz_indu main

https://alz-indu.onrender.com

Your free instance will spin down with inactivity, which can delay requests by 50 seconds or more.

June 22, 2026 at 3:32 PM Building

db100c6 Changement URL power BI

All logs Search logs

Jun 22

03:37:37 PM [4tdzb] lead to breaking code or invalid results. Use at your own risk. For more info please refer to: https://scikit-learn.org/stable/model_persistence.html#security-maintainability-limitations

03:37:37 PM [4tdzb] warnings.warn()

03:37:37 PM [4tdzb] INFO: Started server process [1]

03:37:37 PM [4tdzb] INFO: Waiting for application startup.

03:37:37 PM [4tdzb] INFO: Application startup complete.

03:37:37 PM [4tdzb] INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)

03:37:38 PM [4tdzb] INFO: 127.0.0.1:35878 - "HEAD / HTTP/1.1" 404 Not Found

03:37:45 PM ==> Your service is live 🎉

03:37:45 PM ==>

03:37:45 PM ==> Available at your primary URL <https://alz-indu.onrender.com>

03:37:45 PM ==>

03:37:45 PM ==>

03:37:45 PM [4tdzb] INFO: 10.192.160.5:0 - "GET / HTTP/1.1" 404 Not Found

Need better ways to work with logs? Try the [Render CLI](#), [Render MCP Server](#), or set up a [log stream integration](#)

Figure 5.3

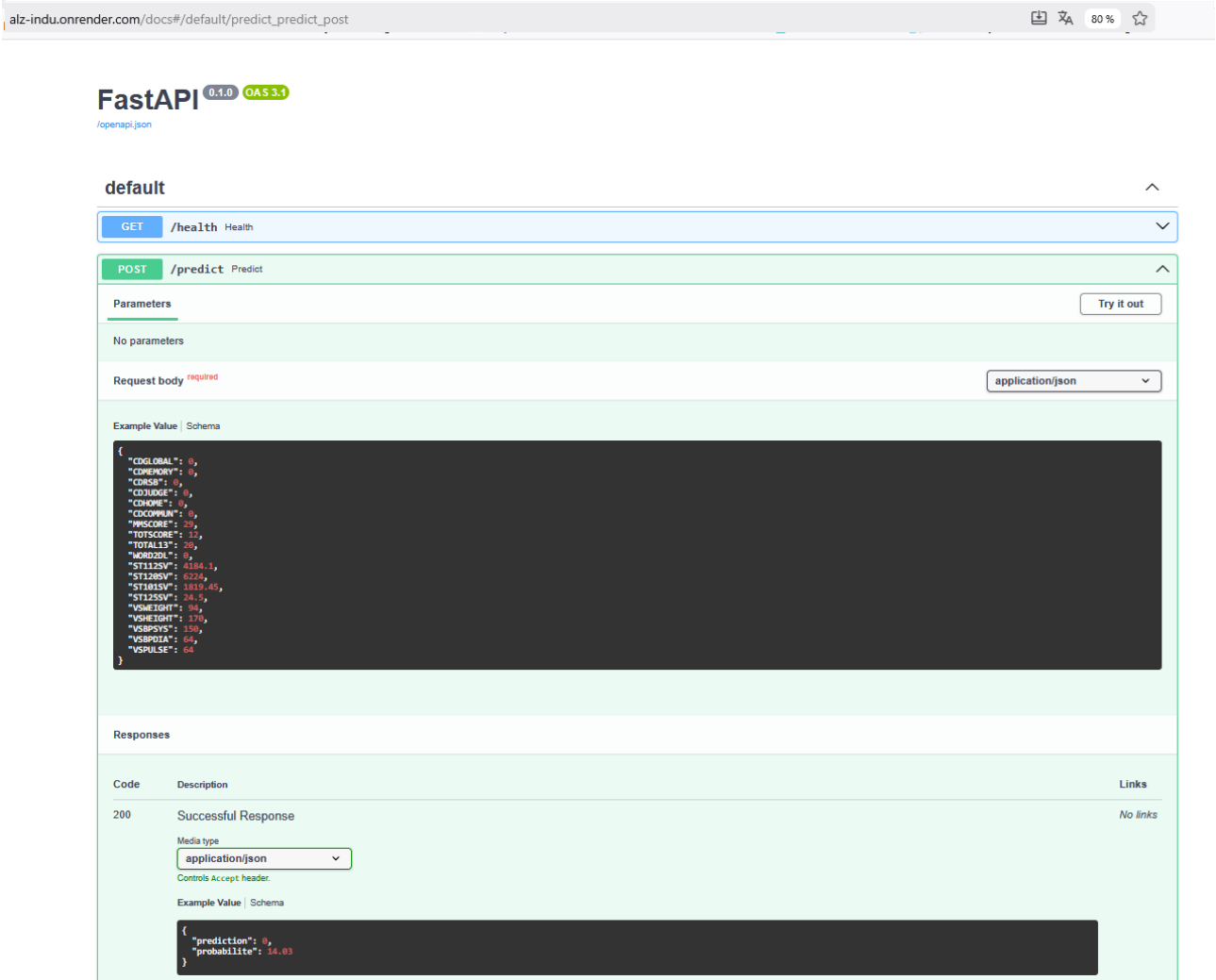


Figure 5.4

dashboard.render.com/web/srv-d8cs776k1jcs73a5jq5g/deployments/dep-d8sjodb6sc1c73cj3hmg?r=2026-06-22%4013%3A43%3A21~2026-06-22%4013%3A47%3A46

Alz_indu_frontend

WEB SERVICE

Alz_indu_frontend

Python 3

Free

Upgrade your instance →

Service ID: `srv-d8cs776k1jcs73a5jq5g`

zoesarini / Alz_indu main

<https://alz-indu-frontend.onrender.com>

ⓘ

Your free instance will spin down with inactivity, which can delay requests by 50 seconds or more.

June 22, 2026 at 3:44 PM

Live

`db100c6` [Changement URL power BI](#)

All logs

Search logs

Jun 22

03:46:42 PM [9hbnh]

03:46:45 PM [9hbnh] 2026-06-22 13:46:45.319 Uvicorn server started on 0.0.0.0:10000

03:46:46 PM ==> Your service is live 🎉

03:46:46 PM ==>

03:46:46 PM ==> //////////////////////////////////////

03:46:46 PM ==>

03:46:46 PM ==> Available at your primary URL <https://alz-indu-frontend.onrender.com>

03:46:46 PM ==>

03:46:46 PM ==> //////////////////////////////////////

03:46:47 PM [9hbnh]

03:46:47 PM [9hbnh] You can now view your Streamlit app in your browser.

03:46:47 PM [9hbnh]

03:46:47 PM [9hbnh] Local URL: <http://localhost:10000>

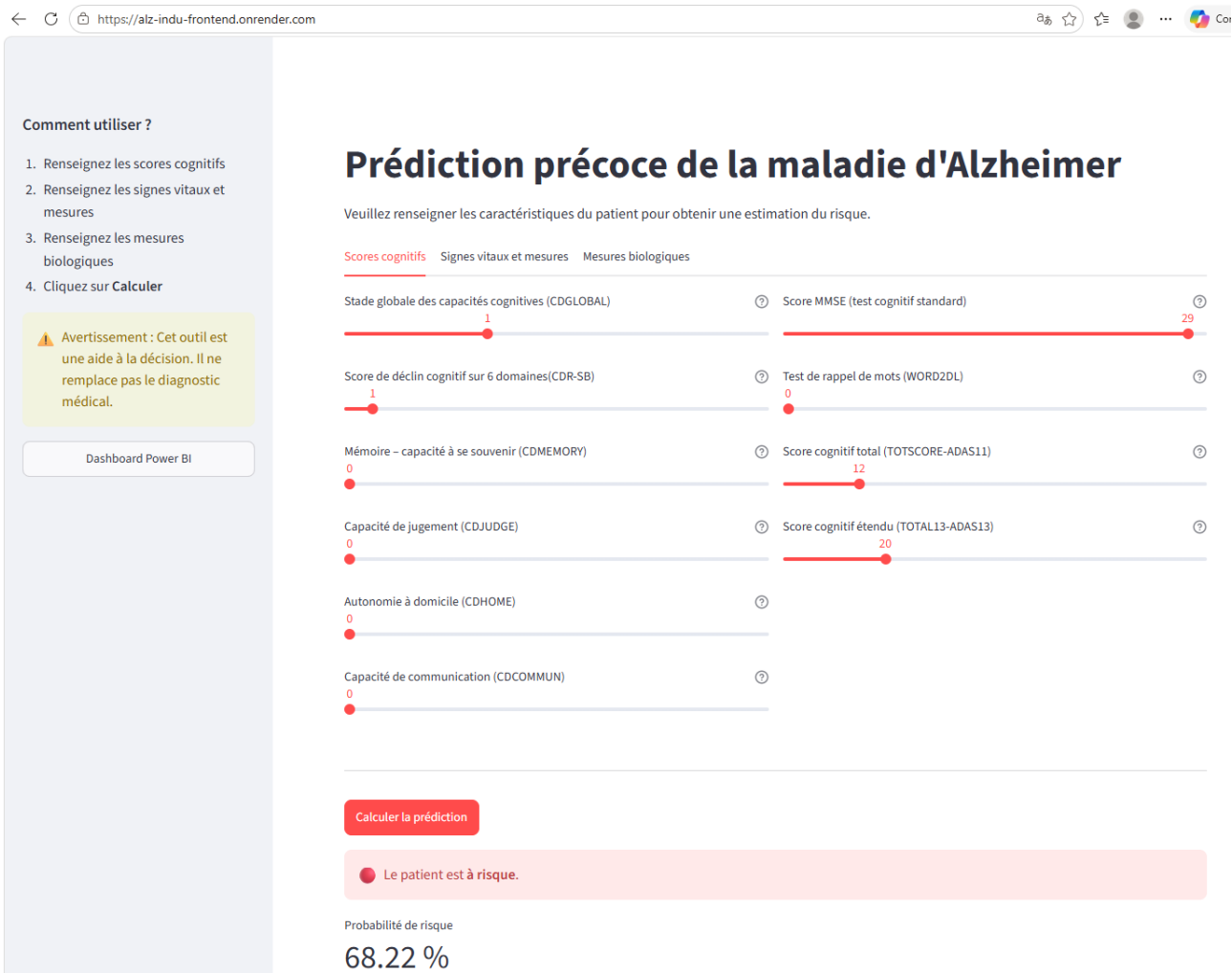
03:46:47 PM [9hbnh] Network URL: <http://10.193.129.107:10000>

03:46:47 PM [9hbnh] External URL: <http://74.220.51.26:10000>

03:46:47 PM [9hbnh]

Need better ways to work with logs? Try the [Render CLI](#), [Render MCP Server](#), or set up a [log stream integ](#)

Figure 5.5



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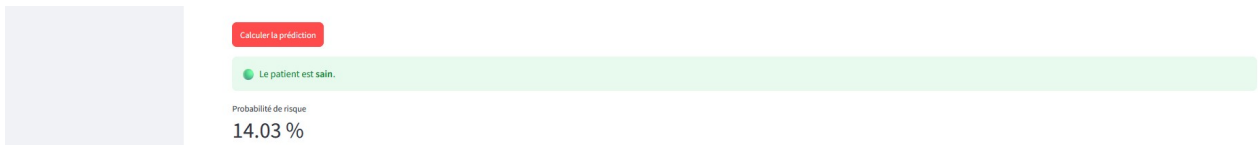
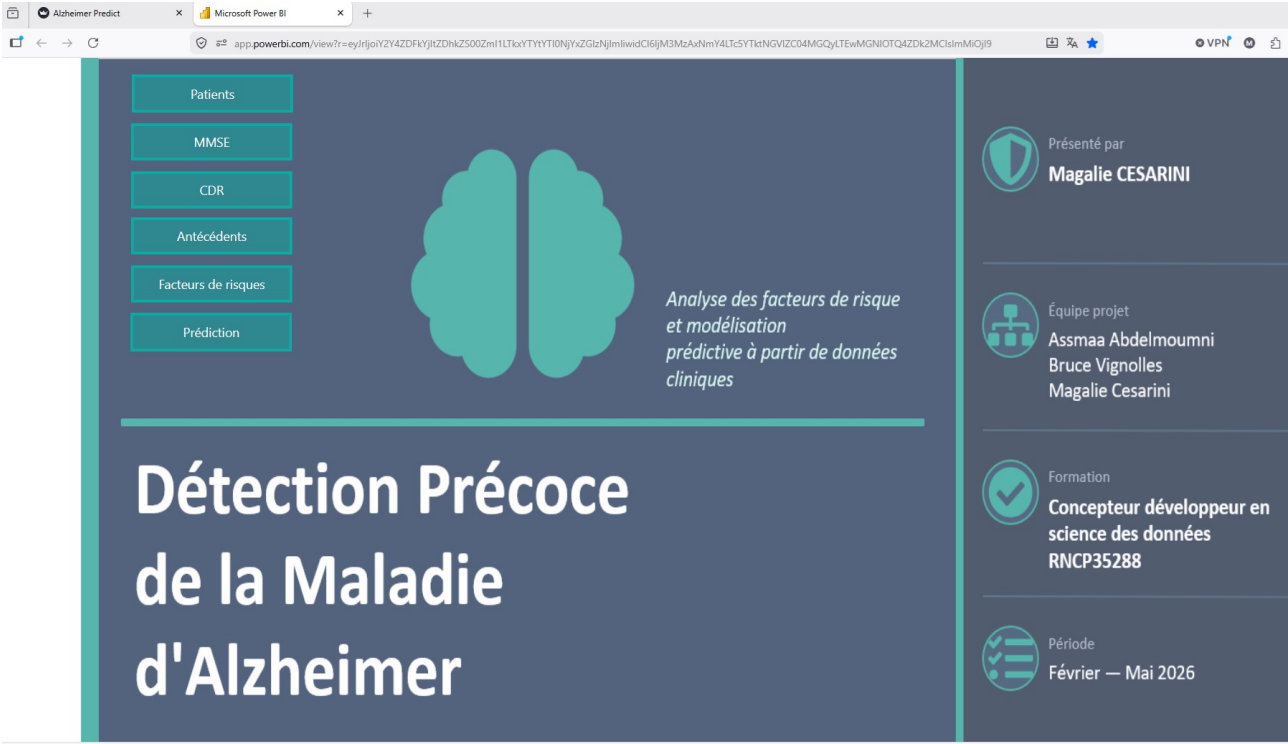


Figure 5.6



Limites

Le projet utilise l'offre gratuite Render. Une montée en charge nécessiterait une infrastructure plus robuste dans un contexte de production.